

STEP 1: Import Required Libraries

```
import numpy as np
import pandas as pd

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
```

STEP 2: Load Dataset

```
data = load_breast_cancer()

X = pd.DataFrame(data.data, columns=data.feature_names)
y = data.target

print(X.head())
```

```
   mean radius  mean texture  mean perimeter  mean area  mean smoothness  \
0         17.99         10.38         122.80      1001.0         0.11840
1         20.57         17.77         132.90      1326.0         0.08474
2         19.69         21.25         130.00      1203.0         0.10960
3         11.42         20.38          77.58        386.1         0.14250
4         20.29         14.34         135.10      1297.0         0.10030

   mean compactness  mean concavity  mean concave points  mean symmetry  \
0          0.27760          0.3001          0.14710          0.2419
1          0.07864          0.0869          0.07017          0.1812
2          0.15990          0.1974          0.12790          0.2069
3          0.28390          0.2414          0.10520          0.2597
4          0.13280          0.1980          0.10430          0.1809

   mean fractal dimension  ...  worst radius  worst texture  worst perimeter  \
0          0.07871  ...         25.38         17.33         184.60
1          0.05667  ...         24.99         23.41         158.80
2          0.05999  ...         23.57         25.53         152.50
3          0.09744  ...         14.91         26.50          98.87
4          0.05883  ...         22.54         16.67         152.20

   worst area  worst smoothness  worst compactness  worst concavity  \
0        2019.0          0.1622          0.6656          0.7119
1        1956.0          0.1238          0.1866          0.2416
2        1709.0          0.1444          0.4245          0.4504
3         567.7          0.2098          0.8663          0.6869
4        1575.0          0.1374          0.2050          0.4000

   worst concave points  worst symmetry  worst fractal dimension
0          0.2654          0.4601          0.11890
1          0.1860          0.2750          0.08902
2          0.2430          0.3613          0.08758
3          0.2575          0.6638          0.17300
4          0.1625          0.2364          0.07678
```

```
[5 rows x 30 columns]
```

STEP 3: Identify Feature Types

```
numeric_features = X.columns.tolist()
categorical_features = []
```

STEP 4: Create Preprocessing Using ColumnTransformer

```
preprocessor = ColumnTransformer(
    transformers=[
        ("num", StandardScaler(), numeric_features)
    ]
)
```

STEP 5: Create Complete ML Pipeline

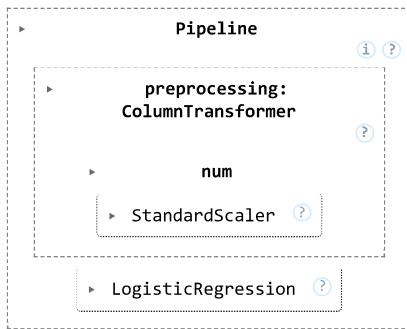
```
pipeline = Pipeline(  
    steps=[  
        ("preprocessing", preprocessor),  
        ("model", LogisticRegression(max_iter=2000))  
    ]  
)
```

STEP 6: Train-Test Split

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y,  
    test_size=0.2,  
    random_state=42,  
    stratify=y  
)
```

STEP 7: Train the Pipeline

```
pipeline.fit(X_train, y_train)
```



STEP 8: Make Predictions

```
y_pred = pipeline.predict(X_test)
```

STEP 9: Evaluate Model Performance

```
print("Accuracy:", accuracy_score(y_test, y_pred))  
print("Precision:", precision_score(y_test, y_pred))  
print("Recall:", recall_score(y_test, y_pred))  
print("F1 Score:", f1_score(y_test, y_pred))
```

```
Accuracy: 0.9824561403508771  
Precision: 0.9861111111111112  
Recall: 0.9861111111111112  
F1 Score: 0.9861111111111112
```

